

Explainer: Problem 5 (Rudin 7.3) — Products Don't Preserve Uniform Convergence

The challenge

Build two sequences $\{f_n\}$, $\{g_n\}$ where:

- $f_n \rightarrow f$ uniformly on some set E ,
- $g_n \rightarrow g$ uniformly on E ,
- BUT $f_n g_n \rightarrow fg$ only *pointwise*, not uniformly.

Context (from Problem 7.2): If both sequences are uniformly bounded, uniform convergence *is* preserved by products. So your counterexample must involve at least one **unbounded** sequence.

Why unboundedness breaks things

Uniform convergence means $\sup_x |f_n(x) - f(x)| \rightarrow 0$. For the product:

$$f_n g_n - fg = (f_n - f)g_n + f(g_n - g).$$

The first term is small in sup-norm *only if* g_n is bounded. If $g_n(x)$ grows large somewhere, a tiny error $(f_n - f)$ can be amplified enormously.

Classic recipe

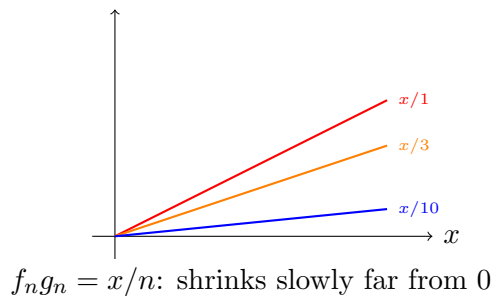
Take $E = \mathbb{R}$ (or any unbounded set).

- $f_n(x) = \frac{1}{n}$ (constants, $\rightarrow 0$ uniformly).
- $g_n(x) = x$ (no n -dependence, $\rightarrow x$ “uniformly” trivially).

Then $f_n g_n = x/n \rightarrow 0$ pointwise, but

$$\sup_{x \in \mathbb{R}} \left| \frac{x}{n} - 0 \right| = +\infty$$

for every n . So the convergence is not uniform.



Alternative recipe

You could also use $f_n(x) = x + 1/n$ (with $f(x) = x$) and $g_n(x) = 1/n$ (with $g = 0$): then $f_n g_n = x/n + 1/n^2$, same issue. The point is to keep one factor *unbounded* while the other shrinks.

Takeaway

Uniform convergence is a statement about the *worst-case* pointwise error, measured in sup-norm. Multiplication by an unbounded function can blow up a uniformly-small error into a uniformly-large error.